

Gregory Macfarlane

Curriculum vitae – Short

Post-graduate employment

Brigham Young University

since 08/2026 Associate Professor, *Civil and Construction Engineering*

11/2018-09/2026 Assistant Professor, *Civil and Construction Engineering*

Transport Foundry

05/2017-11/2018 Data scientist

WSP | Parsons Brinckerhoff

10/2016-05/2017 Technical Principal, *Systems Analysis Group*

05/2014-10/2016 Data scientist, *Systems Analysis Group*

Higher education

Georgia Institute of Technology

2014 PhD, *Civil Engineering*

2014 MS, *Economics*

Brigham Young University

2009 BS, *Civil Engineering*, with University Honors

Publications

† BYU undergraduate student ‡ BYU graduate student

Refereed journal articles

- [15] **Macfarlane, G.**, ‡Stucki, E., Redelfs, A. & Spruance, L. (2026). Where's dinner coming from? A utility-based investigation of food access in Utah. *Journal of Transport and Land Use*, 19(1), 249–277. <https://doi.org/10.5198/jtlu.2026.2554>
- [14] ‡Gibbons, N. & **Macfarlane, G. S.** (2025). Evaluating the Impacts of Parameter Uncertainty in a Practical Transportation Demand Model. *Future Transportation*, 5(1), 15. <https://doi.org/10.3390/futuretransp5010015>
- [13] **Macfarlane, G. S.**, ‡Barnes, M. & †Gibbons, N. (2025). A Utility-Based Approach to Modeling Systemic Resilience of Highway Networks with an Application in Utah. *Journal of Transportation Engineering, Part A: Systems*, 151(1), 04024094. <https://doi.org/10.1061/JTEPBS.TEENG-8534>
- [12] **Macfarlane, G. S.**, ‡Riches, G., ‡Youngs, E. K. & Nielsen, J. A. (2024). Classifying Location Points as Daily Activities using Simultaneously Optimized DBSCAN-TE Parameters.. *Findings*. <https://doi.org/10.32866/001c.116197>
- [11] ‡Wang, B., Fulda, N., †Huang, Z., Schultz, G. G., **Macfarlane, G. S.**, †Arnesen, J. & †Khayyat, A. (2024). Predicting Directional Traffic Volume at Intersections with Automated Traffic Signal Performance Measures Data Using Machine Learning Algorithms. *Transportation Research Record*, 03611981241252829. <https://doi.org/10.1177/03611981241252829>

- [10] ‡Wang, B., Schultz, G. G., **Macfarlane, G. S.**, Eggett, D. L. & †Davis, M. C. (2023). A Methodology to Detect Traffic Data Anomalies in Automated Traffic Signal Performance Measures. *Future Transportation*, 3(4), 1175–1194. <https://doi.org/10.3390/futuretransp3040064>
- [9] **Macfarlane, G. S.**, ‡Stucki, E., Redelfs, A. H. & Spruance, L. A. (2022). Beyond Proximity: Utility-Based Access from Location-Based Services Data. *International Journal of Environmental Research and Public Health*, 19(19), 12352. <https://doi.org/10.3390/ijerph191912352>
- [8] **Macfarlane, G. S.**, Boyd, N., Taylor, J. E. & Watkins, K. (2021). Modeling the impacts of park access on health outcomes: A utility-based accessibility approach. *Environment and Planning B: Urban Analytics and City Science*, 48(8), 2289–2306. <https://doi.org/10.1177/2399808320974027>
- [7] **Macfarlane, G. S.**, †Hunter, C., †Martinez, A. & †Smith, E. (2021). Rider Perceptions of an On-Demand Microtransit Service in Salt Lake County, Utah. *Smart Cities*, 4(2), 717–727. <https://doi.org/10.3390/smartcities4020036>
- [6] Glenn, J., †Bluth, M., †Christianson, M., †Pressley, J., Taylor, A., **Macfarlane, G. S.** & Chaney, R. A. (2020). Considering the Potential Health Impacts of Electric Scooters: An Analysis of User Reported Behaviors in Provo, Utah. *International Journal of Environmental Research and Public Health*, 17(17), 6344. <https://doi.org/10.3390/ijerph17176344>
- [5] Brakewood, C., **Macfarlane, G. S.** & Watkins, K. (2015). The impact of real-time information on bus ridership in New York City. *Transportation Research Part C: Emerging Technologies*, 53, 59–75. <https://doi.org/10.1016/j.trc.2015.01.021>
- [4] **Macfarlane, G. S.**, Garrow, L. A. & Moreno-Cruz, J. (2015). Do Atlanta residents value MARTA? Selecting an autoregressive model to recover willingness to pay. *Transportation Research Part A: Policy and Practice*, 78, 214–230. <https://doi.org/10.1016/j.tra.2015.05.010>
- [3] **Macfarlane, G. S.**, Garrow, L. A. & Mokhtarian, P. L. (2015). The influences of past and present residential locations on vehicle ownership decisions. *Transportation Research Part A: Policy and Practice*, 74, 186–200. <https://doi.org/10.1016/j.tra.2015.01.005>
- [2] Binder, S., **Macfarlane, G. S.**, Garrow, L. A. & Bierlaire, M. (2014). Associations among household characteristics, vehicle characteristics and emissions failures: An application of targeted marketing data. *Transportation Research Part A: Policy and Practice*, 59, 122–133. <https://doi.org/10.1016/j.tra.2013.11.005>
- [1] Wall, T. A., **Macfarlane, G. S.** & Watkins, K. E. (2014). Exploring the Use of Egocentric Online Social Network Data to Characterize Individual Air Travel Behavior. *Transportation Research Record*, 2400(1), 78–86. <https://doi.org/10.3141/2400-09>

Peer-reviewed conference papers

- [4] **Macfarlane, G. S.**, ‡Day, C., Erhardt, G. D. & Watkins, K. E. (2026). Identifying ridehail riders through pairing ActivitySim and BEAM. In *Procedia Computer Science* (pp. 286–293). <https://doi.org/10.1016/j.procs.2026.04.038>
- [3] ‡Atchley, H., †Mansfield, K. W. & **Macfarlane, G. S.** (2025). A Comparative Illustration of Trip- and Activity-Based Modeling Techniques. In *International Conference on Transportation and Development 2025* (pp. 12–23). American Society of Civil Engineers. <https://doi.org/10.1061/9780784486207.002>
- [2] ‡Jarvis, D. L., **Macfarlane, G. S.**, †Woolley, B. & Schultz, G. G. (2024). Simulating Incident Management Team Response and Performance. In *Procedia Computer Science* (pp. 91–96). <https://doi.org/10.1016/j.procs.2024.06.002>
- [1] **Macfarlane, G. S.** & ‡Lant, N. (2023). How Far Are We From Transportation Equity? Measuring the Effect of Wheelchair Use on Daily Activity Patterns. In C. Antoniou, F. Busch, A. Rau & M. Hariharan (Eds.),

Proceedings of the 12th International Scientific Conference on Mobility and Transport (pp. 141–155). Springer Nature Singapore. https://doi.org/10.1007/978-981-19-8361-0_10

Reports

- [2] Macfarlane, G. S. & Atchley, S. H. (2023). *Identifying Microtransit Service Areas Through Microsimulation*. UT-23.01, Utah Department of Transportation. rosap.nntl.bts.gov/view/dot/66312
- [1] Schultz, G. G., Macfarlane, G. S., Wang, B. & Davis, M. C. (2022). *Detecting Traffic Data Anomalies in Longitudinal Signal Performance Measures*. US-22.21, Utah Department of Transportation. <https://rosap.nntl.bts.gov/view/dot/65833>

Conferences & invited talks

Conferences

P: paper/presentation, M: moderator, C: organizing committee

- 2026 hEART European Association for Research in Transportation Annual Symposium^P, 17th International Conference on Ambient Systems Networks and Technologies (ANT – Istanbul)^P
- 2025 Elsevier Transportation Research Symposium^P, Modeling Mobility (MoMo)^{P, M}, International Conference on Transportation & Development (ICTD)^P
- 2024 Activity-based Modeling Symposium (Munich)^{P, C}, International Conference on Travel Behavior Research (IATBR – Vienna)^P, 15th International Conference on Ambient Systems Networks and Technologies (ANT – Hasselt)^P
- 2023 Activity-based modeling symposium (Austin)^P, Innovations in Transportation Analysis and Planning (Indianapolis)^{P, M}
- 2022 Mobil.TUM (Singapore)^P

Invited talks

- 2025 California HSR model transferability, CalTrans Transit Research-to-Practice Symposium
- 2023 Modeling with Big Data, course presentation, Technische Universität München
- 2020 Using Mobile Data to Improve Transportation Sustainability, Utah Central ASCE Chapter luncheon

Research funding

External funding

Principal investigator (\$598,000 total)

- 2026 Macfarlane, G.S. and Brown, N.. *Building automated processes for baseline employment data*. \$65,000, Utah Department of Transportation.
- 2025 Macfarlane, G.S. and Atchley, S.H.. *Establishing Performance Measures for Transportation and Quality of Life*. \$68,000, Utah Department of Transportation.
- 2022 Macfarlane, G.S. and Brown, N.M.. *Activity-based Model Implementation and Analysis Considerations*. \$70,000, Utah Department of Transportation.

- 2021 Macfarlane, G.S., Redelfs, A.H., and Spruance, L.A.. *Equitable Access to Nutrition in Utah*. \$70,000, Utah Department of Transportation.

Co-principal investigator (\$1.44 million total, \$320,000 to BYU)

- 2024 Watkins, K.E. (PI), Erhardt, G.D., and Macfarlane, G.S.. *The Potential for Behavioral Change in New High Speed Rail Lines*. \$280,000, California High Speed Rail Authority and Deutsche Bahn.
- 2020 Watkins, K.E. (PI), Hunter, M.S., Van Hentenryck, P., Peeta, S., Brakewood, C., Cherry, C., Erhardt, G.D., and Macfarlane, G.S.. *T-SCORE: Transit Serving Communities Optimally, Responsibly, and Efficiently*. \$1,000,000, United States Department of Transportation.

Courses

CCE 102: Sustainable Infrastructure

Sustainability in built environments and infrastructure with a focus on global challenges and solutions. Converted from CCE 201 in Fall 2024.

CE 361: Introduction to Transportation Engineering

Transportation systems characteristics, traffic engineering and operations, transportation planning, geometric design, pavement design, transportation safety, freight, public transport, sustainable transportation.

CE 565: Urban Transportation Planning

Characteristics of urban transportation planning and decision making, intermodal transportation, land-use transportation interrelationships, transportation demand modeling, site impact analysis, sustainable transportation, and livable cities.

CE 662: Transport Simulation and Analysis

An advanced graduate course in traffic flow theory and simulation. Topics include shock wave analysis, discrete event simulation of queues and daily activity pattern choices, car following models, and traffic simulation. Laboratory assignments use MATSim and PTV Vissim simulation software.

Awards & honors

- 2023 **TUM Global Visiting Professor**. Selected to enrich the research culture at Technische Universität München through innovative approaches and exploration of new, cutting-edge research fields.
- 2022 **Most Influential Faculty**. Given to the Civil Engineering faculty member whom graduating seniors named as the most influential on their undergraduate education.
- 2022 **ASCE ExCEED Teaching Fellow**. Participated in a week-long intensive teacher development program.

Citizenship & service

External citizenship

- 2023–present **NCHRP 08-184**: Peer review panel member. Framework for Assessing Induced Demand Effects of Various Roadway Investments
- 2019–2025 **Transportation Research Board, AEP50: Travel Demand Forecasting**: Committee member; chair, travel forecasting resources subcommittee; editor of tfresource.org

Professional memberships

2022–present American Society of Civil Engineers
2020–2022 Zephyr Foundation
2009 Tau Beta Pi. Utah β '09

Internal citizenship

2023–present **Department undergraduate committee:** Co-chair. Member since 2021. Lead curriculum revisions for the civil engineering program and ABET accreditation and continuous-improvement efforts.